FBISE | SSC-II | CLASS 10

# COMPUTER SCIENCE
# SLO-Based Question Bank

Possible short and long questions
Summative for Theory only

**241 short questions + 99 long questions**
All 17 theory SLOs across 8 content domains.

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## Scope and use

This practice bank follows the assessment labels in the supplied National Curriculum of Pakistan 2022-23, Grade X Computer Science assessment framework (13 pages). It includes only SLOs explicitly labelled **Summative for Theory**.

No formative or PBA-only SLO is included. Practical HTML/CSS tasks, JavaScript implementation and debugging tasks, and business-pitch production are excluded. Language-independent algorithms and written database tasks remain because they are explicitly included in theory SLOs.

Questions use simple English and include definitions, comparisons, reasoning, scenarios, calculations, algorithms and interpretation. No solutions are included.

These are possible practice questions, not an official FBISE paper or a guarantee of what will appear. The suggested practice marks are 2 for short questions and 5 for long questions. Brackets show a proposed split of the 5 marks; actual examination allocations may differ.

# Coverage and study guide

Each domain below maps to the exact theory SLO codes used in the supplied framework. The final domain uses the EN prefix printed in that source.

| SLO / domain | Topic | Short | Long |
|---|---|---|---|
| CS-10-A-01 | Number systems and data representation | 18 | 6 |
| CS-10-A-02 | Operating systems | 18 | 7 |
| CS-10-A-03 | Software and development tools | 15 | 6 |
| CS-10-B-01 | Counting, logic, searching and sorting | 21 | 9 |
| CS-10-B-02 | Abstraction, modular solutions and trace tables | 12 | 6 |
| CS-10-C-01 | Front-end and back-end development | 10 | 4 |
| CS-10-D-01 | Data science, AI and machine learning | 15 | 5 |
| CS-10-D-02 | Data visualization and relational databases | 18 | 10 |
| CS-10-D-03 | Applying the data science life cycle | 14 | 6 |
| CS-10-E-01 | IoT, Blockchain and cloud computing | 13 | 6 |
| CS-10-E-02 | Applications of AI | 14 | 4 |
| CS-10-E-03 | Social and ethical implications of AI | 9 | 4 |
| CS-10-F-01 | Responsible internet use and information security | 15 | 6 |
| CS-10-F-02 | Cyberbullying, cybercrime and protection | 14 | 6 |
| CS-10-F-03 | The digital divide and accessible solutions | 10 | 4 |
| CS-10-G-01 | Communicating ideas through digital media | 11 | 4 |
| EN-10-H-01 | Digital market and customer research | 14 | 6 |

## Excluded SLOs

**CS-10-C-02, C-03, C-04 and C-05:** practical HTML/CSS, JavaScript and testing/debugging work.
**EN-10-H-02:** pitching a business idea. These are marked Summative for Practical Based Assessment in the supplied framework. No separate formative-labelled SLO was found in this file.

## How to practise

For short questions, write a direct definition, two relevant points, or brief working as requested. For long questions, answer every part, use headings and show complete calculations or trace tables. Treat every question as independent.

## Source note

Primary scope: the supplied theory/formative assessment-framework PDF, pages 1-13. Question style was informed by the previously provided complete solved-paper compilation: Model Paper 2025, Second Annual 2026 and First Annual 2026 Papers B, D and M. Earlier-paper questions outside the present theory scope have not been carried over.

THEORY SLO 01 / 17 | CS-10-A-01

# Number systems and data representation

**Included scope:** Number systems; text encoding; binary arithmetic; signed values; instruction sets and addressing modes.

## Short questions | suggested 2 marks each

**S001.** Define a bit and a byte. State the relationship between them.

**S002.** Why do computers represent data in binary? Give two reasons.

**S003.** State the base and allowed digits of binary, octal, decimal and hexadecimal number systems.

**S004.** Convert (45)10 into binary. Show your working.

**S005.** Convert (101101)2 into decimal. Show the place values used.

**S006.** Convert (7B)16 into binary and octal.

**S007.** Convert (156)8 into decimal and binary.

**S008.** Add (1011)2 and (0110)2. Show the carries.

**S009.** Subtract (0101)2 from (1100)2 using binary subtraction.

**S010.** Find the 1's complement and 2's complement of the 8-bit number 00101101.

**S011.** Differentiate between signed and unsigned binary numbers.

**S012.** Write the range of an 8-bit unsigned integer and an 8-bit signed 2's-complement integer.

**S013.** What is overflow? Explain using the 4-bit unsigned addition 1111 + 0001.

**S014.** What is underflow in the storage of very small non-zero real numbers?

**S015.** Define an encoding scheme. Why are standard character codes needed?

**S016.** Give two differences between ASCII and Unicode.

**S017.** The ASCII code of A is decimal 65. Write its 8-bit binary code.

**S018.** Define an instruction set and an addressing mode.

# CS-10-A-01 | Number systems and data representation

## Long questions | suggested 5 marks each

**L001.** Convert (3A7)16 into binary and octal. Show each conversion step. [2+3]

**L002.** Represent +25 and -25 in 8-bit 2's-complement form. Use 2's-complement addition to calculate 42 - 25 and interpret the result. [2+3]

**L003.** Explain binary addition and overflow. Calculate 0111 + 0001 as 4-bit unsigned numbers and as 4-bit signed 2's-complement numbers; explain why the interpretations differ. [1+2+2]

**L004.** Explain how text is represented using character codes. Compare ASCII and Unicode, and explain why Unicode is useful for English and Urdu text. [2+2+1]

**L005.** Explain data representation, an instruction set and addressing modes. Give a simple example of an instruction that uses an immediate value and one that uses a memory address. [1+2+2]

**L006.** Explain bits, bytes and signed and unsigned numbers. State the unsigned and signed 2's-complement ranges for 8 bits and explain the difference. [1+2+2]

THEORY SLO 02 / 17 | CS-10-A-02

# Operating systems

**Included scope:** OS services; process and thread management; scheduling; synchronization; interrupts; file systems; memory and virtual memory.

## Short questions | suggested 2 marks each

**S019.** Define an operating system. Give two examples.

**S020.** State two resources managed by an operating system.

**S021.** How does an operating system help an application communicate with hardware?

**S022.** What is a system call? Give one example of a service requested through it.

**S023.** Differentiate between a process and a thread.

**S024.** What is process scheduling? Why is it required?

**S025.** What is synchronization? Give one situation where it is needed.

**S026.** Define deadlock and give a simple example involving two resources.

**S027.** What is an interrupt? Why is it useful?

**S028.** State two functions of file management.

**S029.** What is a file system? Explain the purpose of folders.

**S030.** State two functions of memory management.

**S031.** Differentiate between main memory and virtual memory.

**S032.** Why may a computer slow down when many applications are open?

**S033.** Differentiate between single-user and multi-user operating systems.

**S034.** What is a real-time operating system? Give two applications.

**S035.** How do device drivers help with input and output?

**S036.** State the main stages in a process's life, from creation to completion.

# CS-10-A-02 | Operating systems

## Long questions | suggested 5 marks each

**L007.** Define an operating system. Explain its file management and memory management functions with examples. [1+2+2]

**L008.** Explain how an operating system manages input, output, storage, processing and running applications. [5]

**L009.** Explain process scheduling and synchronization. Describe a deadlock involving a printer and scanner. [2+1+2]

**L010.** Describe process states: new, ready, running, waiting and terminated. Show possible transitions in a labelled diagram. [3+2]

**L011.** Compare batch, time-sharing and real-time operating systems. Give a suitable use of each. [3+2]

**L012.** A student runs a browser, word processor and video editor together. Explain CPU scheduling, RAM allocation and virtual memory in this situation. Suggest two ways to improve performance. [3+2]

**L013.** Explain how a program written in a programming language runs on computer hardware, referring to translation, the operating system and the processor's instruction set. [5]

THEORY SLO 03 / 17 | CS-10-A-03

# Software and development tools

**Included scope:** Software categories; productivity software; translators; editors; IDEs; debuggers; repositories; build systems; online, offline and hosted software.

## Short questions | suggested 2 marks each

**S037.** Differentiate between system software and application software, with examples.

**S038.** What is driver software? Give one example.

**S039.** What is programming software? Name two tools used by programmers.

**S040.** Differentiate between a compiler and an interpreter.

**S041.** What is an assembler? What kind of language does it translate?

**S042.** Define an IDE. Name two tools it usually contains.

**S043.** Differentiate between a source-code editor and a debugger.

**S044.** What is a code repository? State one benefit of using it.

**S045.** How does version control help a team of programmers? Give two points.

**S046.** What is a build system? State its purpose.

**S047.** Differentiate between on-premises and cloud-hosted applications.

**S048.** Give two benefits and two limitations of online computing platforms.

**S049.** State two situations where an offline application is useful.

**S050.** Name suitable productivity software for writing a report and analysing numerical data.

**S051.** A new printer is connected but does not work properly. Which type of software may be needed, and why?

## Long questions | suggested 5 marks each

**L014.** Describe five software-development tools: a translator, editor, debugger, IDE and code repository. State the purpose of each. [5]

**L015.** Compare system, application, programming and driver software. Give one example of each and explain how they work together on a computer. [4+1]

**L016.** Compare compilers, interpreters and assemblers, referring to their input, purpose and way of translation. [5]

**L017.** Explain how code repositories and version control support change tracking, recovery and teamwork. [2+3]

**L018.** A school must choose between locally installed and cloud-hosted productivity applications. Compare internet needs, access, collaboration, maintenance and control of data. [5]

**L019.** Explain the roles of a language editor, compiler, debugger and build system when developing a program. Describe how an IDE combines such tools. [4+1]

THEORY SLO 04 / 17 | CS-10-B-01

# Counting, logic, searching and sorting

**Included scope:** Counting principles; algorithm properties; Boolean, verbal and non-verbal reasoning; linear and binary search; bubble and insertion sort.

## Short questions | suggested 2 marks each

**S052.** State the addition principle of counting and give an example.

**S053.** A student has 3 shirts and 4 trousers. How many different shirt-trouser pairs are possible? Name the counting principle.

**S054.** Differentiate between a permutation and a combination.

**S055.** In how many ways can 4 different books be arranged on a shelf?

**S056.** In how many ways can a team of 2 be chosen from 5 students?

**S057.** State the pigeonhole principle. Apply it to 13 students and 12 birth months.

**S058.** In a class, 18 students like Python, 12 like JavaScript and 5 like both. How many like at least one of these languages?

**S059.** Define an algorithm. State any two properties of a good algorithm.

**S060.** Differentiate between definiteness and finiteness.

**S061.** What is meant by the effectiveness and generality of an algorithm?

**S062.** Find the value of (P AND NOT Q) OR Q when P is true and Q is false.

**S063.** Differentiate between AND, OR and NOT using simple statements.

**S064.** All tablets are computing devices. Some computing devices are portable. Can we conclude that all tablets are portable? Give a reason.

**S065.** Complete the pattern 2, 4, 8, 16, __, __ and state the rule.

**S066.** A shape pattern repeats: circle, square, triangle. Which shape is in position 8? Explain.

**S067.** Differentiate between searching and sorting.

**S068.** Describe linear search. Can it work on unsorted data?

**S069.** State two conditions or requirements for efficient binary search on an array.

**S070.** Why is binary search usually faster than linear search on a large sorted array?

**S071.** What happens during one pass of bubble sort?

**S072.** Briefly describe how insertion sort builds a sorted list.

# CS-10-B-01 | Counting, logic, searching and sorting

## Long questions | suggested 5 marks each

**L020.** Explain six properties of a good algorithm: input, output, definiteness, finiteness, effectiveness and generality. Include a simple example. [3+2]

**L021.** Compare linear and binary search. Find 35 in [5, 10, 15, 20, 25, 30, 35] using both methods and show the comparisons. [2+3]

**L022.** Use bubble sort to arrange [7, 3, 5, 2, 4] in ascending order. Show the list after each full pass and explain the stopping condition. [4+1]

**L023.** Use insertion sort to arrange [8, 4, 6, 2, 5] in ascending order. Show the list after each insertion and explain the method. [4+1]

**L024.** A club has 6 students. Calculate the ways to choose a captain and vice-captain, and the ways to choose a 2-student team. Explain why the answers differ. [2+2+1]

**L025.** In a group, 20 students study art, 15 study music and 8 study both. Find the number studying at least one subject. Then explain why among 13 students at least two share a birth month. [2+3]

**L026.** Construct truth tables for P AND Q, P OR Q and NOT P. Use them to evaluate (P OR Q) AND NOT Q for all input pairs. [3+2]

**L027.** Write a language-independent algorithm to find a student's roll number in an unsorted list. Explain why your chosen search method is suitable. [3+2]

**L028.** A password consists of one uppercase English letter followed by three decimal digits. Repetition is allowed. Calculate the number of passwords. Then calculate the total if the three digits cannot repeat. Explain the principles used. [2+2+1]

THEORY SLO 05 / 17 | CS-10-B-02

# Abstraction, modular solutions and trace tables

**Included scope:** Algorithm design; generalization; modularity; dry runs; syntax and logic errors. Use language-independent pseudocode.

## Short questions | suggested 2 marks each

**S073.** Define abstraction. Give one everyday example.

**S074.** How does abstraction help create a generalized solution?

**S075.** Give two advantages of dividing a solution into modules.

**S076.** Differentiate between an algorithm and a program.

**S077.** What is a dry run? Why is it useful?

**S078.** What information is recorded in a trace table?

**S079.** Differentiate between a syntax error and a logic error.

**S080.** A programmer calculates a rectangle's area as length + width. Identify the error and correct the formula.

**S081.** Trace this pseudocode: total = 0; FOR i = 1 TO 3: total = total + i; END FOR. What is the final total?

**S082.** A loop intended to run 5 times uses i < 5, starting at i = 1 and increasing by 1. How many times does it run, and how can it be corrected?

**S083.** Why should a reusable average-calculation module accept a list and its size rather than fixed marks for one class?

**S084.** Name two modules that could be used in a student-result system.

## Long questions | suggested 5 marks each

**L029.** Write pseudocode to input five numbers, find their sum and mean, and display both. Trace it for 2, 4, 6, 8 and 10 using a table. [3+2]

**L030.** Write a generalized algorithm to find the largest of n numbers. Dry-run it for [12, 7, 19, 5] and show the largest value after each step. [3+2]

**L031.** Design a modular solution for a student-result system using input, calculation and output modules. Explain how abstraction and reuse help the design. [3+2]

**L032.** Trace the algorithm: x = 1; total = 0; WHILE x <= 4: total = total + x*x; x = x + 1; END WHILE; OUTPUT total. Record x and total during each iteration and state the output. [4+1]

**L033.** An algorithm is: total = 0; count = 1; WHILE count <= 5: INPUT mark; total = mark; END WHILE; mean = total / 4; OUTPUT mean. Identify three logic errors, rewrite it correctly and trace it for marks 10, 20, 30, 40, 50. [2+2+1]

**L034.** Write pseudocode to count even and odd numbers in a list. Use a trace table for [3, 8, 10, 7, 5]. [3+2]

THEORY SLO 06 / 17 | CS-10-C-01

# Front-end and back-end development

**Included scope:** Conceptual roles of browser-side and server-side development only; no practical HTML/CSS or JavaScript implementation tasks.

## Short questions | suggested 2 marks each

**S085.** Differentiate between front-end and back-end development.

**S086.** Name three front-end technologies and state the purpose of each.

**S087.** Name two technologies commonly used for back-end development.

**S088.** Where do front-end and back-end programs run?

**S089.** Give two examples of tasks handled by the back end of an online shop.

**S090.** Give one example of a website feature that can work entirely in the browser. Explain why it may not need a back end.

**S091.** Why does a website with user accounts usually require a back end?

**S092.** Describe the roles of the front end and back end when a user logs in.

**S093.** Can JavaScript be used on both the front end and back end? Explain briefly.

**S094.** Why should important checks, such as permission to access an account, be enforced on the server?

## Long questions | suggested 5 marks each

**L035.** Compare front-end and back-end development in terms of purpose, execution location, technologies and tasks. Explain how they work together in an online shop. [3+2]

**L036.** Describe the roles of HTML, CSS and browser JavaScript in the front end. Explain how a back-end program and database support a student-result website. [3+2]

**L037.** A user logs into a school portal to view results. Explain the request and response process from the browser to the server and back. Draw a labelled conceptual diagram. [3+2]

**L038.** Compare a browser-only calculator with a website that stores users' orders. Explain why their back-end requirements differ. [2+3]

THEORY SLO 07 / 17 | CS-10-D-01

# Data science, AI and machine learning

**Included scope:** Definitions and relationships; supervised, unsupervised and reinforcement learning; classification, regression, churn and segmentation.

## Short questions | suggested 2 marks each

**S095.** Define data science and state one use.

**S096.** Define Artificial Intelligence and give one example.

**S097.** Define Machine Learning and explain its relationship with AI.

**S098.** State two differences between data science and Machine Learning.

**S099.** Differentiate between supervised and unsupervised learning.

**S100.** What is labelled data? Give an example.

**S101.** Differentiate between classification and regression.

**S102.** Is predicting next month's sales a classification or regression problem? Give a reason.

**S103.** Is dividing customers into groups using buying habits supervised or unsupervised learning? Explain.

**S104.** What is reinforcement learning? Give one simple example.

**S105.** How can Machine Learning help detect fraudulent transactions?

**S106.** What is churn prediction? How can a business use its results?

**S107.** What is behavioural segmentation? Give one benefit.

**S108.** State two benefits of automation and scalability in Machine Learning.

**S109.** A model labels customers as high or low risk of leaving. What does a high-risk prediction mean, and what should the business do?

## Long questions | suggested 5 marks each

**L039.** Define data science, AI and Machine Learning. Compare their purposes and explain how they overlap using one real-world example. [3+2]

**L040.** Compare supervised, unsupervised and reinforcement learning. Explain each with a suitable example. [3+2]

**L041.** Explain classification, regression and clustering. Choose a suitable type for spam detection, sales prediction and customer grouping, giving reasons. [2+3]

**L042.** Explain how a company can use churn prediction and behavioural segmentation. Describe how the results can guide different actions for customers. [2+3]

**L043.** Describe the data science life cycle from collecting data to interpreting model outcomes. Use a fraud-detection example. [5]

THEORY SLO 08 / 17 | CS-10-D-02

# Data visualization and relational databases

**Included scope:** Charts; centre and spread; outliers; plot interpretation; relational models, keys, schema and written database queries.

## Short questions | suggested 2 marks each

**S110.** State two benefits of data visualization.

**S111.** Differentiate between a bar chart and a histogram.

**S112.** When is a scatter plot useful? State what its two axes represent.

**S113.** What does a boxplot show? Name two features visible in it.

**S114.** What is a dotplot? When is it useful?

**S115.** Why are side-by-side bar charts useful for comparing groups?

**S116.** Calculate the mean and median of 2, 4, 4, 6 and 9.

**S117.** Find the range of 12, 15, 18, 20 and 25. What does it describe?

**S118.** What is an outlier? How can it affect the mean?

**S119.** What is meant by the shape, centre and spread of a dataset?

**S120.** A school's pass percentages are 70, 75, 80 and 85 over four years. Describe the trend and choose a suitable chart.

**S121.** A scatter plot shows higher sales on hotter days. Why does this alone not prove that temperature is the only cause?

**S122.** What is a relational database? Define an entity and an attribute.

**S123.** Differentiate between a primary key and a foreign key.

**S124.** How does a database table differ from a database form or report?

**S125.** State the purpose of SELECT, CREATE TABLE and JOIN in SQL.

**S126.** How can data stored in a database be used in Machine Learning?

**S127.** Write a SELECT query to display Name and Marks from Student where Marks >= 80.

# CS-10-D-02 | Data visualization and relational databases

## Long questions | suggested 5 marks each

**L044.** For marks [10, 12, 12, 14, 17], calculate the mean, median, mode and range. Replace 17 with 50 and explain the effect on the mean and median. [3+2]

**L045.** A shop sells 20, 35, 25 and 40 notebooks in January, February, March and April. Draw a labelled bar chart. State two observations and explain which chart would best emphasize change over time. [3+1+1]

**L046.** Choose suitable visualizations for subject enrolments, grouped heights, study hours versus marks, and the distribution of marks including outliers. Explain each choice. [4+1]

**L047.** Compare bar charts, histograms, scatter plots, dotplots and boxplots. Give one suitable use of each. [5]

**L048.** Two classes have marks A = [40, 50, 60] and B = [10, 50, 90]. Calculate their means and ranges. Compare their centres and spreads and explain why one average does not describe all differences. [3+2]

**L049.** A school stores Student(StudentID, Name, ClassID) and Class(ClassID, ClassName). Identify entities, attributes, primary keys and the foreign key. Draw the one-to-many E-R relationship and write the relational schema. [2+2+1]

**L050.** Using Student(StudentID, Name, ClassID) and Class(ClassID, ClassName), write a query to list student names with class names using JOIN. Explain the joining field and the purpose of a primary key. [3+2]

**L051.** Write SQL to create Student with StudentID as an integer primary key, Name as text and Marks as an integer. Then write a query to show names and marks for students scoring at least 80, ordered by marks from highest to lowest. [3+2]

**L052.** Explain logical and physical database design. For a library with Book and Member entities, add a Loan entity to record borrowing, identify suitable keys and explain the relationships. [2+3]

**L053.** Monthly sales are 100, 120, 140 and 160 units. Advertising also rises during these months. Describe the trend, suggest a useful plot and explain why seasonal demand could be a confounding factor. [2+1+2]

THEORY SLO 09 / 17 | CS-10-D-03

# Applying the data science life cycle

**Included scope:** Problem formulation; data sources, collection, storage and preparation; model choice; interpreting results; mean squared error.

## Short questions | suggested 2 marks each

**S128.** State the main stages of the data science life cycle.

**S129.** Turn the business question "What will our sales be next month?" into a data science problem.

**S130.** What data could help predict a store's sales? Give two examples.

**S131.** State two sources of data for a data science project.

**S132.** Why must collected data be cleaned before modelling?

**S133.** Give two examples of formatting or preparing data for analysis.

**S134.** What is model building in the data science life cycle?

**S135.** Name a problem suitable for simple linear regression.

**S136.** What is a decision tree? State one possible use.

**S137.** State two factors to consider when selecting a model.

**S138.** What is the purpose of evaluating a predictive model?

**S139.** Write the formula for Mean Squared Error (MSE). What does a smaller MSE indicate when comparing the same prediction task?

**S140.** Calculate MSE when actual values are [2, 4] and predicted values are [3, 3].

**S141.** Why should predictions be interpreted in the context of the original business problem?

## Long questions | suggested 5 marks each

**L054.** A shop wants to predict next year's sales from the previous three years. Explain problem definition, data collection, storage, preparation, model building and evaluation. [5]

**L055.** A company says, "Our customers are leaving us." Formulate a data science problem, suggest useful data, choose a suitable learning approach and explain how predictions can guide action. [1+1+1+2]

**L056.** Calculate MSE for actual values [3, 5, 7, 9] and predicted values [2, 6, 6, 10]. Show errors, squared errors and the final average in a table. Explain the purpose of this metric. [4+1]

**L057.** Model A predicts [11, 18, 31] and Model B predicts [12, 19, 29] for actual values [10, 20, 30]. Calculate each MSE, identify the better result on these data and justify your choice. [2+2+1]

**L058.** Explain how to gather data from a database, remove duplicate records, handle missing values and format the data for modelling. Use a simple school or shop example. [5]

**L059.** Compare simple linear regression and a decision tree at an introductory level. Suggest a suitable model for predicting sales and one for classifying customers as likely or unlikely to leave. Give reasons. [3+2]

THEORY SLO 10 / 17 | CS-10-E-01

# IoT, Blockchain and cloud computing

**Included scope:** Foundational IoT components; applications and scenario suitability; Blockchain uses; cloud benefits.

## Short questions | suggested 2 marks each

**S142.** Define the Internet of Things (IoT). Give one example.

**S143.** Name three foundational components of an IoT system.

**S144.** What is the role of a sensor in IoT?

**S145.** How do connectivity and data processing support IoT?

**S146.** Explain how a soil-moisture sensor can help a farmer.

**S147.** A thermometer displays a temperature but cannot communicate it. Is it automatically an IoT device? Give a reason.

**S148.** Define Blockchain. State one application.

**S149.** How does linking blocks with hashes help detect changes?

**S150.** Give two applications of Blockchain other than cryptocurrency.

**S151.** What is a shared or distributed ledger?

**S152.** State two benefits of cloud computing.

**S153.** How can cloud computing support an IoT system?

**S154.** Why can Blockchain not guarantee that a faulty sensor's original reading is true?

## Long questions | suggested 5 marks each

**L060.** Explain how IoT works through sensors, connectivity and data processing. Use smart irrigation as an example and draw a labelled diagram. [3+2]

**L061.** Explain the use of IoT in healthcare, farming, homes, transport and industry. [5]

**L062.** Describe how Blockchain records and validates transactions. Explain its use in cryptocurrency and tracking assets. [3+2]

**L063.** Explain how IoT and Blockchain can work together to record medicine temperatures during transport. Describe the benefit and one limitation of this arrangement. [3+1+1]

**L064.** A school needs shared storage, access from home and room to store more data later. Explain five benefits of using cloud computing for these needs. [5]

**L065.** Compare an ordinary offline appliance with an IoT appliance. Explain the added roles of sensors, network connection, processing and action in a smart system. [1+4]

THEORY SLO 11 / 17 | CS-10-E-02

# Applications of AI

**Included scope:** Speech recognition; personal assistants; Quran memorization support; NLP applications; robotics.

## Short questions | suggested 2 marks each

**S155.** Define speech recognition and give one application.

**S156.** How can a personal assistant use speech recognition?

**S157.** Explain one way AI can support a Quran memorization application.

**S158.** Differentiate between speech recognition and speech-to-text output.

**S159.** How can voice-based authentication use speech information?

**S160.** Define Natural Language Processing (NLP). Give one example.

**S161.** How does NLP help in email filtering?

**S162.** State one use each of language translation and document analysis.

**S163.** What is predictive text? Give a daily-life example.

**S164.** What is sentiment analysis? What can it show about customer reviews?

**S165.** Give two applications of robotics in manufacturing.

**S166.** How can robots help in search and rescue operations?

**S167.** State one use each of robots in homes and farming.

**S168.** How can speech recognition support security-related work? State one privacy concern.

## Long questions | suggested 5 marks each

**L066.** Describe five applications of speech recognition, including personal assistants, speech-to-text, Quran memorization support, authentication and security-related use. [5]

**L067.** Explain the use of NLP in spam filtering, language translation, document analysis, predictive text and sentiment analysis. [5]

**L068.** Explain applications of robotics in manufacturing, search and rescue, home cleaning and agriculture. Describe one benefit of using robots in a dangerous task. [4+1]

**L069.** For each task, identify a suitable AI area and explain its role: typing spoken words, translating a message, classifying reviews, finding trapped people and answering a spoken question. [5]

THEORY SLO 12 / 17 | CS-10-E-03

# Social and ethical implications of AI

**Included scope:** Unfair outcomes; social injustice; responsibility of designers; ethical norms and human benefit.

## Short questions | suggested 2 marks each

**S169.** What is AI bias? Give one simple example.

**S170.** How can unrepresentative training data produce unfair decisions?

**S171.** State two ethical responsibilities of AI designers.

**S172.** Why is human review important for high-impact AI decisions?

**S173.** Give two ways AI can benefit society.

**S174.** How can improper use of AI harm a group of people?

**S175.** Why should an AI system protect personal data?

**S176.** What does transparency mean when an organization uses AI?

**S177.** A speech-recognition system works poorly for one regional accent. What fairness problem does this show, and how could it be reduced?

## Long questions | suggested 5 marks each

**L070.** Explain how AI can cause social injustice. Use two examples and suggest measures to make AI decisions fairer. [3+2]

**L071.** Describe five ethical principles for designing AI systems that benefit people. [5]

**L072.** A recruitment system repeatedly rejects suitable applicants from one community. Explain possible causes, effects on people and steps the organization should take. [2+1+2]

**L073.** Discuss responsible AI use in public services or law enforcement, referring to fairness, privacy, accuracy, human review and accountability. [5]

THEORY SLO 13 / 17 | CS-10-F-01

# Responsible internet use and information security

**Included scope:** Technology addiction; online information; privacy; spam, spyware and cookies; cloud security; protection methods and the purpose of technology laws.

## Short questions | suggested 2 marks each

**S178.** What is technology addiction? State two possible effects.

**S179.** Give two habits that support balanced use of digital devices.

**S180.** Define cybersecurity. Why is it needed?

**S181.** Differentiate between privacy and data security.

**S182.** What is spam? How can it affect a user?

**S183.** What is spyware? State one risk it creates.

**S184.** What are browser cookies? Give one useful purpose and one privacy concern.

**S185.** What is phishing? State one way to avoid it.

**S186.** Give two ways the internet can be used to share helpful information.

**S187.** How can false information harm people? State one way to check it.

**S188.** Give two ways to protect information stored in the cloud.

**S189.** Differentiate between authentication and authorization.

**S190.** How do encryption and software updates protect data or devices?

**S191.** What is the purpose of digital privacy and cybercrime laws?

**S192.** How can social-media peer pressure lead to unsafe sharing?

## Long questions | suggested 5 marks each

**L074.** Explain five strategies for safe and responsible internet use that protect personal information. [5]

**L075.** Compare spam, spyware, phishing and tracking through cookies. Explain one protective action for each. [4+1]

**L076.** Explain the effects of technology addiction on study, health and relationships. Suggest a balanced-use plan for a student. [3+2]

**L077.** Compare passwords, multi-factor authentication, encryption, firewalls and software updates as methods of protecting devices and data. [5]

**L078.** Explain key security and privacy concerns of cloud computing. Suggest measures involving account security, access permissions, backups and careful sharing. [2+3]

**L079.** Explain how technology laws can support digital privacy and information security. Discuss responsible information sharing and respect for others' personal data. [3+2]

THEORY SLO 14 / 17 | CS-10-F-02

# Cyberbullying, cybercrime and protection

**Included scope:** Harassment impacts and response; backups; access control; monitoring; public and private clouds; hardware/software security; reporting concepts.

## Short questions | suggested 2 marks each

**S193.** Define cyberbullying. Give one example.

**S194.** State two effects of cyberbullying on a student.

**S195.** What should a victim do before deleting abusive messages?

**S196.** Give two ways to support a friend who is being harassed online.

**S197.** State two actions schools or parents can take to prevent cyberbullying.

**S198.** Why should users block and report abusive accounts?

**S199.** What is a backup? Why is it needed?

**S200.** How do access permissions protect files?

**S201.** Give two ways network monitoring can help detect threats.

**S202.** Differentiate between public and private clouds.

**S203.** Give one hardware method and one software method used to protect a device.

**S204.** What is a cybercrime? Give two examples.

**S205.** State the general steps for reporting an online offence while keeping evidence safe.

**S206.** Why should users avoid replying aggressively or forwarding abusive content?

## Long questions | suggested 5 marks each

**L080.** A student receives repeated threatening messages. Explain a safe response involving evidence, blocking, platform reporting, support from a trusted adult and formal reporting where necessary. [5]

**L081.** Evaluate the roles of schools and parents in preventing cyberbullying. Suggest five practical measures. [5]

**L082.** Explain how backups, access control, network monitoring, antivirus software and a firewall protect information. [5]

**L083.** Compare public and private clouds in ownership or access, control and suitability. Explain why security measures are necessary in both. [3+2]

**L084.** Describe common cybercrimes and the purpose of laws addressing them. Explain general reporting steps and the types of evidence a victim should preserve. [2+1+2]

**L085.** Prepare a device-protection plan using both hardware and software methods. Explain five measures for a school computer laboratory. [5]

THEORY SLO 15 / 17 | CS-10-F-03

# The digital divide and accessible solutions

**Included scope:** Unequal access; effects on critical information; inclusive application design; ways to reduce inequality; students' role.

## Short questions | suggested 2 marks each

**S207.** Define the digital divide.

**S208.** State two causes of unequal access to digital technology.

**S209.** How can poor internet access affect a student's education?

**S210.** Give two ways the digital divide affects access to important information.

**S211.** Why should an application work on low-cost devices?

**S212.** State two features that help users with slow internet connections.

**S213.** How can local-language content improve access to information?

**S214.** What is the role of accessibility features such as captions or screen-reader support?

**S215.** Give two actions students can take to reduce the digital divide.

**S216.** Why should designers consider users' devices, connectivity and abilities?

## Long questions | suggested 5 marks each

**L086.** Explain causes and effects of the digital divide in education, employment and access to essential information. Suggest two solutions. [3+2]

**L087.** A learning application must serve students with old phones and limited mobile data. Suggest five design features that make it more accessible. [5]

**L088.** Discuss how schools, communities and students can help reduce unequal access to technology and information. [5]

**L089.** Compare an application designed only for fast internet with one offering text, low-data access, local languages, captions and offline content. Explain how these choices affect inclusion. [2+3]

THEORY SLO 16 / 17 | CS-10-G-01

# Communicating ideas through digital media

**Included scope:** Researching simple topics; extracting key ideas; copy and graphics; audience and format; YouTube, Instagram and Facebook; effective publication.

## Short questions | suggested 2 marks each

**S217.** What is digital communication? Give two examples of tools used for it.

**S218.** Why should a message be adapted to its audience?

**S219.** What is meant by extracting the key ideas from information?

**S220.** State two features of effective copy for an event poster.

**S221.** How can graphics make a message easier to understand?

**S222.** Choose a suitable platform for a long teaching video and give a reason.

**S223.** Choose a suitable platform for a product-photo collection and give a reason.

**S224.** How can a community use a Facebook group or page to share information?

**S225.** State two good practices for reaching an audience on digital platforms.

**S226.** Why should creators check accuracy and credit sources before publishing?

**S227.** How can captions and readable text improve digital communication?

## Long questions | suggested 5 marks each

**L090.** A school is announcing a science exhibition. Explain how to collect accurate details, choose key information, write clear copy and select suitable graphics for a digital poster. [5]

**L091.** Compare YouTube, Instagram and Facebook as tools for communicating ideas. Give a suitable audience and type of content for each. [3+2]

**L092.** Explain how the same study tip can be presented as a short video, an image post and a community announcement. Describe how wording and design should change for each format. [3+2]

**L093.** Describe five good practices for publishing clear and responsible digital content, including accuracy, audience awareness, suitable media, readability and crediting sources. [5]

THEORY SLO 17 / 17 | EN-10-H-01

# Digital market and customer research

**Included scope:** Quantitative and qualitative methods; digital research tools; question design; collecting and interpreting results; sampling bias; correlation and causation.

## Short questions | suggested 2 marks each

**S228.** What is market research? Why is it useful before starting a business?

**S229.** Differentiate between quantitative and qualitative research.

**S230.** Give one example of a quantitative question and one qualitative question for an online tutoring service.

**S231.** Name two digital tools for collecting customer opinions.

**S232.** When is an interview more useful than a closed-question survey?

**S233.** What is sample-selection bias? Give one example.

**S234.** Why may surveying only your friends give misleading results?

**S235.** Rewrite this biased question neutrally: "How much do you love our excellent learning app?"

**S236.** Differentiate between an open-ended and a closed-ended research question.

**S237.** What is the difference between correlation and causation?

**S238.** A survey of 50 students finds that 30 prefer evening tuition. What percentage prefer evening tuition?

**S239.** How can social-media comments and reviews provide customer insights?

**S240.** Why should research questions be clear and focused on one issue?

**S241.** How can survey results help a business choose a price or service time?

## Long questions | suggested 5 marks each

**L094.** Plan digital market research for an affordable online tutoring service. State the research aim, target respondents, suitable tools, one quantitative question and one qualitative question. [1+1+1+2]

**L095.** Compare quantitative and qualitative research. Explain when each is useful and how the two can be combined to understand customer needs. [3+2]

**L096.** A business surveys only urban students who already own expensive phones. Explain the sampling bias and suggest ways to obtain a more representative sample for a nationwide study app. [2+3]

**L097.** A survey of 100 students finds that 60 prefer evening lessons, 25 prefer afternoon and 15 prefer morning. Present the percentages, state two useful insights and identify one limitation of these findings. [2+2+1]

**L098.** Students who use an education app have higher marks in a survey. Explain why this does not prove the app caused the higher marks. Suggest two other possible factors and a more careful way to investigate. [2+2+1]

**L099.** Design four unbiased questions to investigate customer needs for a stationery delivery service. Include both quantitative and qualitative questions and explain how one response could guide a business decision. [4+1]